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Exercise for
Embedded Systems

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Sheet 1: Embedded Systems and Microcontrollers

Exercise 1: Questions

- a) What is an embedded system?
- b) What is the source of requirements for an embedded system?
- c) Embedded systems can be categorized into two groups.
 - (1) What are these categories?
 - (2) What are their characteristics?
 - (3) What kind of hardware is typically used for these categories?
 - (4) Which programming languages are most dominant in these categories?
 - (5) Name at least one example for each category!
- d) What is a microcontroller?

Exercise 2: Digital I/O

For this and the following exercises we refer to the Atmel ATmega16 microcontroller. A datasheet can be found in the Moodle room.

Assume 8 buttons are connected to Port A and GND. Also assume 8 LEDs are connected to Port B and VCC (using fitting resistors) such that they can be lit.

- a) What are the registers that control these ports?
- b) How should these registers be initialized?

- c) Write a loop that allows control over the LEDs via the buttons:
 - (1) On a 1-to-1 basis (pushing button 4 causes LED 4 to be lit).
 - (2) Priority encoder: show the binary coding for the number of the highest button pushed.
- d) What is bouncing? Implement a debouncing method.

Exercise 3: Interrupts and Polling

- a) Choose Interrupts or Polling for the following scenarios and explain your choice.
 - (1) The "change input"-button on a monitor
 - (2) The wireless-reciever of a garage-opener
 - (3) The keyboard on a standard desktop
 - (4) The temperature-sensor of a weather-station
- b) When is an ISR called and how is it done?

Exercise 4: Timers and Counters

- a) What is a counter? What is a timer?
- b) What components does timer 1 of the ATmega16 have? How are they configured?
- c) How is the reading and writing of a 16 bit value made atomic?
- d) What is a watch dog?
- e) Why might it be necessary to temporarily disable interrupts when reading 16 bit values?

Exercise 5: Analog Devices

- a) What analog devices can be found on a ATmega16?
- b) What is PWM and how does it work?
- c) Sketch a successive approximation converter and explain how it works.
- d) Imagine you only have 1 Ohm Resistors available, which cost 10 cents each. What is the minimum achievable cost for a 4Bit R-2R and a 4Bit binary weighted resistor circuit respectively if you can only combine resistors in serial?
- e) What are the disadvantages of the binary weighted resistor circuit?